Assignment 1

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Contents

1 Exercise 1.3

The weight update rule in (1.3) has the nice interpretation that it moves in the direction of classifying $\mathbf{x}(t)$ correctly.

(a) Show that $y(t)\mathbf{w}^T(t)\mathbf{x}(t) < 0$.

Answer: To show that $y(t)\mathbf{w}^T(t)\mathbf{x}(t) < 0$, consider the definition of misclassification:

- If the classifier $\mathbf{w}(t)$ correctly classified $\mathbf{x}(t)$, we would have
$$\text{sign}\left(\mathbf{w}^T(t)\mathbf{x}(t)\right) = y(t).$$
- However, since $\mathbf{x}(t)$ is misclassified,
$$\text{sign}\left(\mathbf{w}^T(t)\mathbf{x}(t)\right) \neq y(t),$$
which means
$$y(t)\mathbf{w}^T(t)\mathbf{x}(t) < 0.$$

This inequality $y(t)\mathbf{w}^T(t)\mathbf{x}(t) < 0$ confirms that the example $\mathbf{x}(t)$ is indeed misclassified by the weight vector $\mathbf{w}(t)$, as required.

(b) Show that $y(t)\mathbf{w}^T(t+1)\mathbf{x}(t) > y(t)\mathbf{w}^T(t)\mathbf{x}(t)$.
Answer:

(1) **Equation (1.3):** The weight update rule is given by:
$$\mathbf{w}(t+1) = \mathbf{w}(t) + y(t)\mathbf{x}(t)$$
This rule is used to update the weight vector $\mathbf{w}(t)$ at each step t when the example $\mathbf{x}(t)$ is misclassified.

(2) **Substitute $\mathbf{w}(t+1)$ into the Expression:**
- Expanding $y(t)\mathbf{w}^T(t+1)\mathbf{x}(t)$ using the update rule:
$$y(t)\mathbf{w}^T(t+1)\mathbf{x}(t) = y(t)\left(\mathbf{w}^T(t) + y(t)\mathbf{x}^T(t)\right)\mathbf{x}(t)$$
- Distributings and simplifing:
$$y(t)\mathbf{w}^T(t+1)\mathbf{x}(t) = y(t)\mathbf{w}^T(t)\mathbf{x}(t) + y^2(t)\mathbf{x}^T(t)\mathbf{x}(t)$$

- This further simplifies to:

$$y(t)\mathbf{w}^T(t+1)\mathbf{x}(t) = y(t)\mathbf{w}^T(t)\mathbf{x}(t) + \mathbf{x}^T(t)\mathbf{x}(t)$$

(3) **Simplify the Expression:**

- Since $y^2(t) = 1$ because $y(t)$ can only be $+1$ or -1, so:

$$y(t)\mathbf{w}^T(t+1)\mathbf{x}(t) = y(t)\mathbf{w}^T(t)\mathbf{x}(t) + \mathbf{x}^T(t)\mathbf{x}(t)$$

- The term $\mathbf{x}^T(t)\mathbf{x}(t)$ is the dot product of $\mathbf{x}(t)$ with itself, which is a positive quantity (specifically, it is the squared Euclidean norm of $\mathbf{x}(t)$):

$$\mathbf{x}^T(t)\mathbf{x}(t) > 0$$

- Therefore:

$$y(t)\mathbf{w}^T(t+1)\mathbf{x}(t) = y(t)\mathbf{w}^T(t)\mathbf{x}(t) + \text{positive quantity}$$

(4) **Conclusion:**

- Since we are adding a positive quantity to $y(t)\mathbf{w}^T(t)\mathbf{x}(t)$, we have:

$$y(t)\mathbf{w}^T(t+1)\mathbf{x}(t) > y(t)\mathbf{w}^T(t)\mathbf{x}(t)$$

This shows that the move from $\mathbf{w}(t)$ to $\mathbf{w}(t+1)$ increases the value of $y(t)\mathbf{w}^T(t)\mathbf{x}(t)$, which confirms that the update is indeed moving in the direction that reduces the classification error.

(c) As far as classifying $\mathbf{x}(t)$ is concerned, argue that the move from $\mathbf{w}(t)$ to $\mathbf{w}(t+1)$ is a move "in the right direction".
Answer:

(1) **Classification Improvement:**

- In classification, a weight vector $\mathbf{w}(t)$ correctly classifies an input $\mathbf{x}(t)$ if the sign of $\mathbf{w}^T(t)\mathbf{x}(t)$ matches the label $y(t)$.
- If $\mathbf{x}(t)$ is misclassified by $\mathbf{w}(t)$, then $y(t)\mathbf{w}^T(t)\mathbf{x}(t) < 0$, as shown in part (a).

(2) **Effect of the Update:**

- In part (b), we showed that after updating the weight vector, the value $y(t)\mathbf{w}^T(t+1)\mathbf{x}(t)$ becomes larger than $y(t)\mathbf{w}^T(t)\mathbf{x}(t)$.
- This means that the update increases the alignment between the weight vector and the input vector $\mathbf{x}(t)$ in the sense that the product $y(t)\mathbf{w}^T(t+1)\mathbf{x}(t)$ is closer to being positive (or is positive if the misclassification is corrected).

(3) **Moving in the Right Direction:**

- By moving from $\mathbf{w}(t)$ to $\mathbf{w}(t+1)$, the update is effectively pushing the decision boundary in a way that reduces the error for classifying $\mathbf{x}(t)$.
- Since the update improves the value of $y(t)\mathbf{w}^T(t+1)\mathbf{x}(t)$ compared to $y(t)\mathbf{w}^T(t)\mathbf{x}(t)$, it moves the classification decision for $\mathbf{x}(t)$ in the right direction—towards correct classification.

Conclusion:

The move from $\mathbf{w}(t)$ to $\mathbf{w}(t+1)$ is indeed a move "in the right direction" because it decreases the misclassification error for the input $\mathbf{x}(t)$. This move increases the alignment between the input vector $\mathbf{x}(t)$ and the weight vector, pushing the decision closer to correct classification.

2 Exercise 1.5

Which of the following problems are more suited for the learning approach and which are more suited for the design approach?

(a) Determining the age at which a particular medical test should be performed
Answer: Learning Approach
This problem is better suited for the learning approach because it involves analyzing data from large number of patients to find the optimal age for the test. The data drive nature of this problem is the reason it is more suited for the learning approach.

(b) Classifying numbers into primes and non-primes
Answer: Design Approach
The classification of numbers into primes and non-primes is well-defined and based on mathematical rules. There is no need to learn from data because a precise algorithm can be designed accurately.

(c) Detecting potential fraud in credit card charges
Answer: Learning Approach
Detecting fraud is a complex problem that often involves identifying subtle patterns in transaction data. Fraudsters continually change their tactics, so a system that can learn from new data and adapt over time is crucial.

(d) Determining the time it would take a falling object to hit the ground
Answer: Design Approach
This problem is well-suited for the design approach because it is based on well-established physical laws, such as Newton's laws of motion.

(e) Determining the optimal cycle for traffic lights in a busy intersection
Answer: Learning Approach
This problem could benefit from the learning approach because the optimal cycle for traffic lights can depend on various factors like traffic flow patterns, time of day, and even weather conditions.

3 Exercise 1.6

For each of the following tasks, identify which type of learning is involved (supervised, reinforcement, or unsupervised) and the training data to be used. If a task can fit more than one type, explain how and describe the training data for each type.

(a) Recommending a book to a user in an online bookstore
Answer: Supervised Learning or Reinforcement learning

(1) Supervised Learning: If the system is trained using historical data where users' past preferences (inputs) and their book choices or ratings (outputs) are available, it can learn to predict which books to recommend.

(2) Reinforcement Learning: If the system actively adjusts recommendations based on the user's interactions (clicks, purchases) to maximize engagement or satisfaction, reinforcement learning might be used. The system receives feedback from the user's behavior and improves over time.

(3) Training Data: Historical user data including user profiles, past purchases, ratings, and clicks.

(b) Playing tic-tac-toe
Answer: Reinforcement Learning

(1) Reinforcement Learning: The system learns by playing games, either against itself or others, and adjusting its strategy based on wins, losses, and draws. It receives rewards or penalties based on the outcome of each game.

(2) Training Data: The game states and outcomes from playing numerous games.

(c) Categorizing movies into different types
Answer: Unsupervised Learning

(1) Unsupervised Learning: If the goal is to cluster movies into categories without predefined labels (genres), unsupervised learning is

used. The system might analyze movie attributes like cast, director, plot summary, and viewing patterns to group similar movies together.

(2) Training Data: Data about movies such as plot summaries, actors, genres, and user interaction data (views, ratings) without explicit labels.

(d) Learning to play music
Answer: Reinforcement Learning or Supervised Learning

(1) Reinforcement Learning: If the system is adjusting its playing style based on feedback from listening (like in a self-learning AI that receives feedback on how well it played), reinforcement learning could be applied.

(2) Supervised Learning: If the system is learning from a labeled dataset of notes, sequences, and styles (e.g., training on how humans play), supervised learning would be applicable.

(3) Training Data: For supervised learning, it could be sheet music paired with corresponding performances. For reinforcement learning, feedback signals such as user ratings or specific outcomes from playing.

(e) Credit limit; Deciding the maximum allowed debt for each bank customer
Answer: Supervised Learning

(1) Supervised Learning: The system could be trained using data on customers' profiles (income, spending patterns, credit history) and their corresponding credit limits and repayment behaviors. The model would learn to predict an appropriate credit limit based on these inputs.

(2) Training Data: Historical data on customers including demographics, financial status, past credit limits, and repayment history.

4 Exercise 1.7

For each of the following learning scenarios in the above problem, evaluate the performance of g on the three points in $\mathcal{X}$ outside $\mathcal{D}$. To measure the performance, compute how many of the 8 possible target functions agree with g on all three points, on two of them, on one of them, and on none of them.

(a)
- **Hypothesis Space $\mathcal{H}$**: There are only two hypotheses in $\mathcal{H}$:
  - $h_1(x)$ that always returns '$\bullet$'
  - $h_2(x)$ that always returns '$\circ$'
- **Learning Algorithm**: Picks the hypothesis that matches the data set the most.

Analysis:

- The hypothesis g will match the majority label in the data set $\mathcal{D}$.
- There are 8 possible target functions. Each target function either matches g on all three points, two points, one point, or none.

Count:

- If the majority label in the training set is '$\bullet$', then the hypothesis g will be h_1.
- If the majority label in the training set is '$\circ$', then g will be h_2.
- Given the symmetry of the problem, for 4 target functions g will match all 3 points, for 2 functions it will match 2 points, for another 2 functions it will match 1 point, and for 0 functions it will match none.

(b)
- **Same Hypothesis Space $\mathcal{H}$ as in part (a)**, but the learning algorithm now picks the hypothesis that matches the data set the least.

Analysis:

- This time, g is chosen as the opposite of the majority label in $\mathcal{D}$.
- This means g will perform worst on $\mathcal{D}$.

Count:

- Given the symmetry, g will match none of the points for 4 target functions, 1 point for 2 functions, 2 points for another 2 functions, and 3 points for 0 functions.

(c)
- **Hypothesis Space** $\mathcal{H}$: Contains only the XOR function $h(x) = \text{XOR}(x)$.
- **Learning Algorithm**: Always picks the XOR hypothesis.

Analysis:

- XOR is defined such that $h(x) = \bullet$ if the number of 1's in x is odd, and $h(x) = \circ$ if the number of 1's is even.
- We need to see how XOR aligns with the target function on the 3 points outside $\mathcal{D}$.

Count:

- XOR will match 2 of the 8 possible target functions on all 3 points, 4 of them on 2 points, and 2 of them on 1 point.

(d)
- **Hypothesis Space** $\mathcal{H}$: Contains all possible hypotheses (all Boolean functions on three variables).
- **Learning Algorithm**: Picks the hypothesis that agrees with all training examples but disagrees the most with XOR on the points outside $\mathcal{D}$.

Analysis:

- This approach specifically biases against the XOR pattern.
- g is selected to agree with the training set but to maximize disagreement with XOR on the other points.

Count:

- Since XOR has been targeted for maximum disagreement, g might match 0 or 1 points with XOR for 4 functions, and 2 or 3 points with XOR for the other 4 functions.

Answers:

- **(a)**: 4 target functions match all points, 2 match 2 points, 2 match 1 point, 0 match no points.
- **(b)**: 0 target functions match all points, 2 match 2 points, 2 match 1 point, 4 match no points.
- **(c)**: 2 target functions match all points, 4 match 2 points, 2 match 1 point, 0 match no points.
- **(d)**: 0 target functions match all points, 2 match 2 points, 4 match 1 point, 2 match no points.

5 Problem 1.1

We have 2 opaque bags, each containing 2 balls. One bag has 2 black balls and the other has a black and a white ball. You pick a bag at random and then pick one of the balls in that bag at random. When you look at the ball it is black. You now pick the second ball from that same bag. What is the probability that this ball is also black? *[Hint: Use Bayes' Theorem:* $\mathbb{P}[A \text{ and } B] = \mathbb{P}[A \mid B]\,\mathbb{P}[B] = \mathbb{P}[B \mid A]\,\mathbb{P}[A]$*.]*

Answer:
Let's define the events:

- A_1: The event that Bag 1 is chosen.
- A_2: The event that Bag 2 is chosen.
- B_1: The event that the first ball drawn is black.
- B_2: The event that the second ball drawn is black.

We want to find $\mathbb{P}(B_2 \mid B_1)$, the probability that the second ball is black given that the first ball drawn is black.
By Bayes' Theorem, we have:

$$\mathbb{P}(A_1 \mid B_1) = \frac{\mathbb{P}(B_1 \mid A_1)\mathbb{P}(A_1)}{\mathbb{P}(B_1)}$$

Step 1: Calculate $\mathbb{P}(B_1 \mid A_1)$ **and** $\mathbb{P}(B_1 \mid A_2)$

- If Bag 1 is chosen (which has two black balls):

$$\mathbb{P}(B_1 \mid A_1) = 1$$

- If Bag 2 is chosen (which has one black ball and one white ball):

$$\mathbb{P}(B_1 \mid A_2) = \frac{1}{2}$$

Step 2: Calculate $\mathbb{P}(A_1)$ **and** $\mathbb{P}(A_2)$

- Since the bags are chosen at random:

$$\mathbb{P}(A_1) = \mathbb{P}(A_2) = \frac{1}{2}$$

Step 3: Calculate $\mathbb{P}(B_1)$
$\mathbb{P}(B_1)$ is the total probability of drawing a black ball first, regardless of the bag:

$$\mathbb{P}(B_1) = \mathbb{P}(B_1 \mid A_1)\mathbb{P}(A_1) + \mathbb{P}(B_1 \mid A_2)\mathbb{P}(A_2)$$

$$\mathbb{P}(B_1) = 1 \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

Step 4: Calculate $\mathbb{P}(A_1 \mid B_1)$
Now, using Bayes' Theorem:

$$\mathbb{P}(A_1 \mid B_1) = \frac{\mathbb{P}(B_1 \mid A_1)\mathbb{P}(A_1)}{\mathbb{P}(B_1)} = \frac{1 \times \frac{1}{2}}{\frac{3}{4}} = \frac{2}{3}$$

Similarly,

$$\mathbb{P}(A_2 \mid B_1) = \frac{\mathbb{P}(B_1 \mid A_2)\mathbb{P}(A_2)}{\mathbb{P}(B_1)} = \frac{\frac{1}{2} \times \frac{1}{2}}{\frac{3}{4}} = \frac{1}{3}$$

Step 5: Calculate $\mathbb{P}(B_2 \mid B_1)$
The probability that the second ball is black given that the first ball was black is the probability that we chose Bag 1, plus the probability that we chose Bag 2 and the first ball drawn was black:

$$\mathbb{P}(B_2 \mid B_1) = \mathbb{P}(A_1 \mid B_1) \times \mathbb{P}(B_2 \mid A_1, B_1) + \mathbb{P}(A_2 \mid B_1) \times \mathbb{P}(B_2 \mid A_2, B_1)$$

- If Bag 1 is chosen:
$$\mathbb{P}(B_2 \mid A_1, B_1) = 1$$
- If Bag 2 is chosen:
$$\mathbb{P}(B_2 \mid A_2, B_1) = 0 \quad \text{(since the only other ball in Bag 2 is white)}$$

Thus:

$$\mathbb{P}(B_2 \mid B_1) = \left(\frac{2}{3}\right) \times 1 + \left(\frac{1}{3}\right) \times 0 = \frac{2}{3}$$

Final Answer:
The probability that the second ball is black given that the first ball drawn was black is $\boxed{\frac{2}{3}}$

6 Problem 1.2

Consider the perceptron in two dimensions: $h(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x})$ where $\mathbf{w} = [w_0, w_1, w_2]^\top$ and $\mathbf{x} = [1, x_1, x_2]^\top$. Technically, $\mathbf{x}$ has three coordinates, but we call this perceptron two-dimensional because the first coordinate is fixed at 1.

(a) Show that the regions on the plane where $h(\mathbf{x}) = +1$ and $h(\mathbf{x}) = -1$ are separated by a line. If we express this line by the equation $x_2 = ax_1 + b$, what are the slope a and intercept b in terms of w_0, w_1, w_2?

(b) Draw a picture for the cases $\mathbf{w} = [1, 2, 3]^\top$ and $\mathbf{w} = [-1, 2, 3]^\top$.

In more than two dimensions, the $+1$ and -1 regions are separated by a *hyperplane*, the generalization of a line.

(a) (1) **Perceptron Decision Boundary:**

- The perceptron makes a decision based on the sign of $\mathbf{w}^\top \mathbf{x}$. Specifically:

$$h(\mathbf{x}) = \text{sign}(w_0 + w_1 x_1 + w_2 x_2)$$

- The decision boundary (where $h(\mathbf{x}) = 0$) separates the regions where $h(\mathbf{x}) = +1$ and $h(\mathbf{x}) = -1$. This occurs when:

$$w_0 + w_1 x_1 + w_2 x_2 = 0$$

(2) **Expressing the Line Equation:**

- To express this equation in the form $x_2 = ax_1 + b$, solve for x_2:

$$\begin{aligned} w_2 x_2 &= -w_1 x_1 - w_0 \\ x_2 &= -\frac{w_1}{w_2} x_1 - \frac{w_0}{w_2} \end{aligned}$$

- Here, the slope a and intercept b are given by:

$$a = -\frac{w_1}{w_2}, \quad b = -\frac{w_0}{w_2}$$

So, the line that separates the regions where $h(\mathbf{x}) = +1$ and $h(\mathbf{x}) = -1$ is given by:

$$x_2 = -\frac{w_1}{w_2} x_1 - \frac{w_0}{w_2}$$

(b) **Solution:**

For $\mathbf{w} = [1, 2, 3]^\top$:

- Slope $a = -\frac{w_1}{w_2} = -\frac{2}{3} = -\frac{2}{3}$.
- Intercept $b = -\frac{w_0}{w_2} = -\frac{1}{3}$.

The equation of the line is:

$$x_2 = -\frac{2}{3}x_1 - \frac{1}{3}$$

For $\mathbf{w} = [-1, 2, 3]^\top = [-1, -2, -3]^\top$:

- Slope $a = -\frac{w_1}{w_2} = -\frac{-2}{-3} = -\frac{2}{3}$.
- Intercept $b = -\frac{w_0}{w_2} = -\frac{-1}{-3} = -\frac{1}{3}$.

The equation of the line remains the same:

$$x_2 = -\frac{2}{3}x_1 - \frac{1}{3}$$

Drawing the Picture:

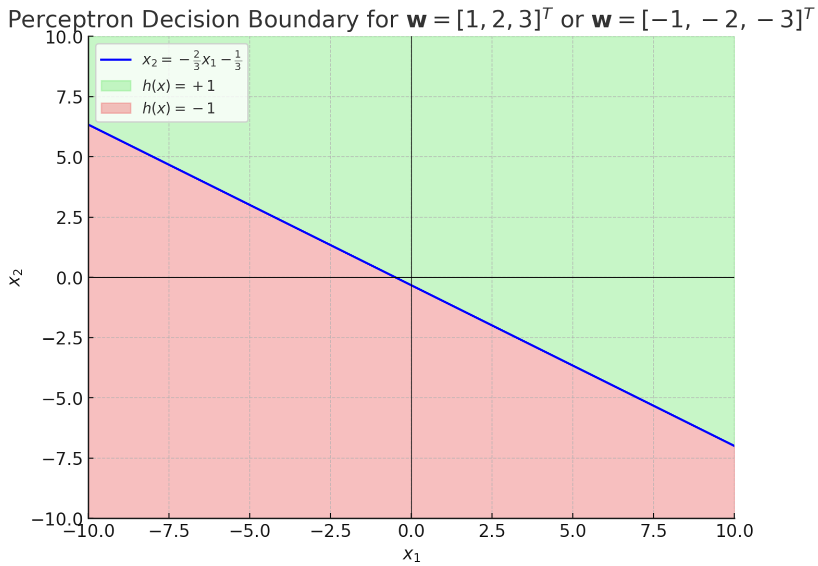

Figure 1: Perceptron Decision Boundary for $\mathbf{w} = [1, 2, 3]^\top$ or $\mathbf{w} = [-1, -2, -3]^\top$

7 Problem 1.4 (a-e)

In Exercise 1.4, we use an artificial data set to study the perceptron learning algorithm. This problem leads you to explore the algorithm further with data sets of different sizes and dimensions.

(a)

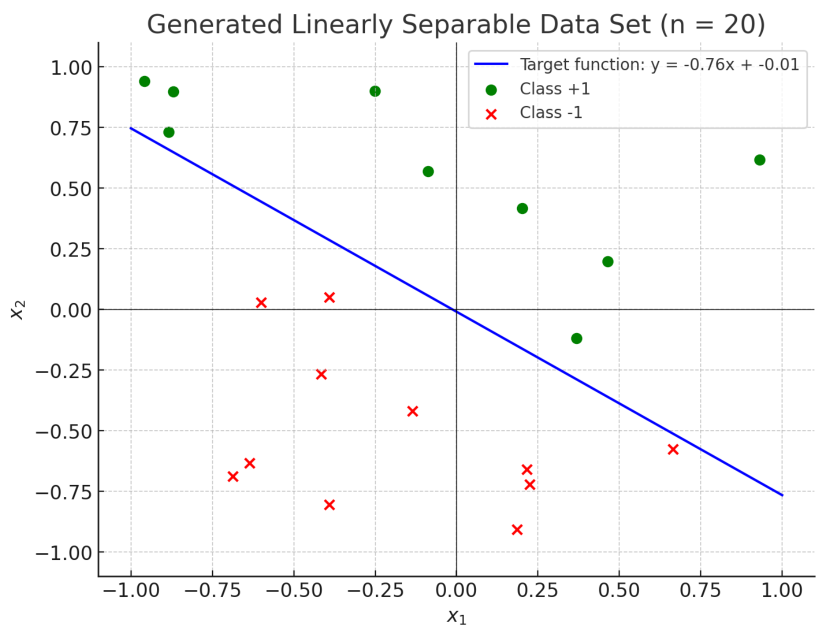

Figure 2: Generated Linearly Seperable Data Set (n=20)

(b) **Commentary:**

- The final hypothesis g closely matches the target function f, as both lines are quite similar. This is expected since the data set is linearly separable, and the perceptron algorithm is guaranteed to converge to a correct classifier in such cases.

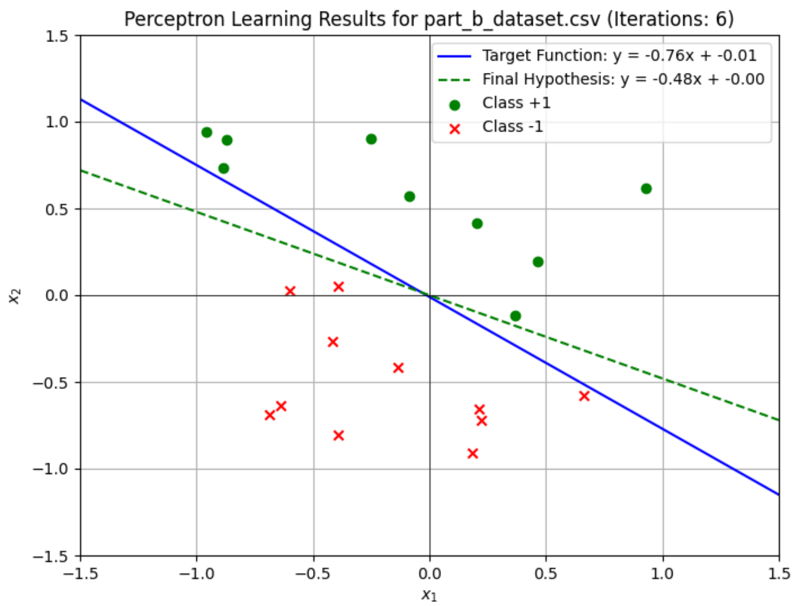

Figure 3: Perceptron Learning Algorithm Results for part b

(c) **Comparison with Part (b):**

- In the first data set, the algorithm converged after just 6 iterations, whereas for this second data set, it took significantly more iterations (133).
- The difference in the number of iterations is likely due to the distribution of points relative to the target function. If the points are closer to the boundary, it may take more updates to correctly classify all points.

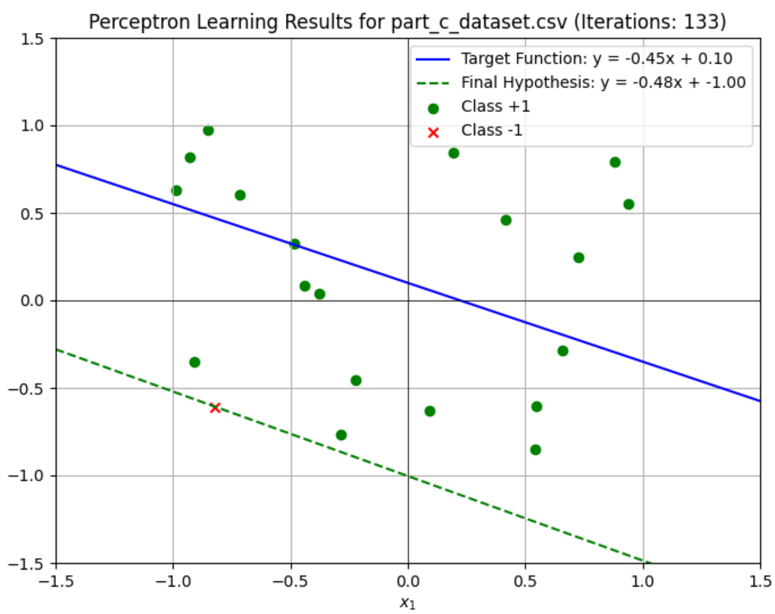

Figure 4: Perceptron Learning Algorithm Results for part c

(d) **Comparison with Part (b):**

- With 100 points, the algorithm took 20 iterations to converge, which is more than the 6 iterations from part (b) but fewer than the 133 iterations in part (c).
- The convergence depends not only on the size of the data set but also on how linearly separable the data points are and how far they are from the decision boundary.

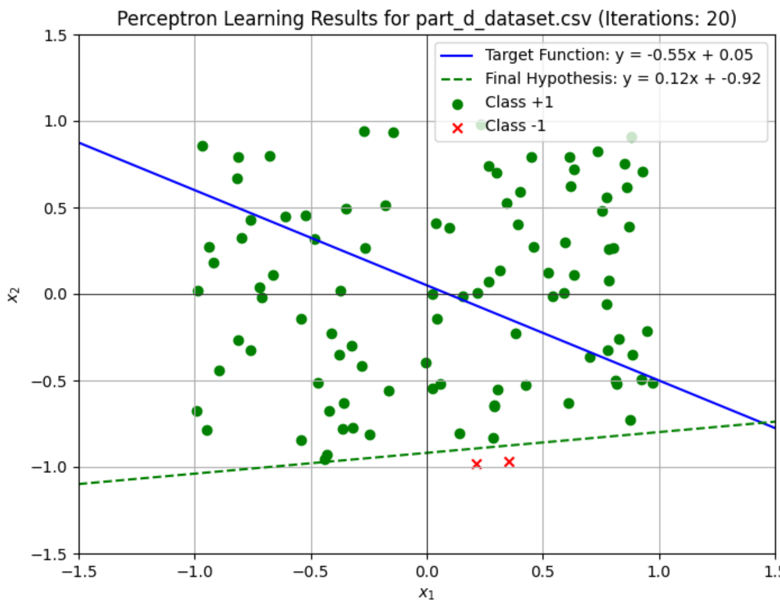

Figure 5: Perceptron Learning Algorithm Results for part d

(e) **Comparison with Part (b):**

- The number of iterations increased significantly compared to the smaller data sets. This is expected as larger data sets tend to have more points near the decision boundary, leading to more updates during training.
- Compared to part (b), where we had only 6 iterations for 20 points, this increase shows the relationship between data size and convergence time for the perceptron algorithm.

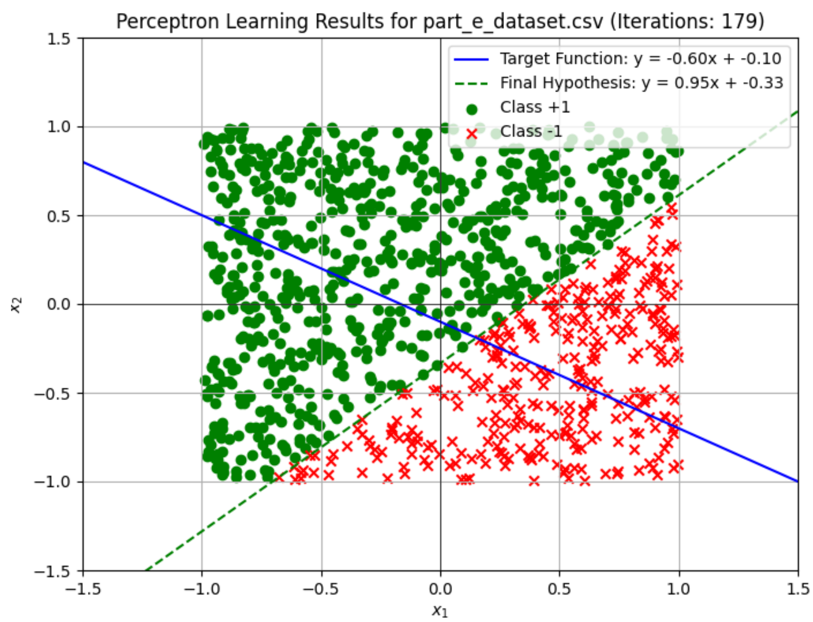

Figure 6: Perceptron Learning Algorithm Results for part e